In [1]:
```
import pandas as pd
dataset=pd.read_csv("Social_Network.csv")
dataset
dataset=pd.get_dummies(dataset,drop_first=True)
print(dataset)
```

```
      User ID  Age  EstimatedSalary  Purchased  Gender_Male
0    15624510   19            19000          0         True
1    15810944   35            20000          0         True
2    15668575   26            43000          0        False
3    15603246   27            57000          0        False
4    15804002   19            76000          0         True
..        ...  ...              ...        ...          ...
395  15691863   46            41000          1        False
396  15706071   51            23000          1         True
397  15654296   50            20000          1        False
398  15755018   36            33000          0         True
399  15594041   49            36000          1        False

[400 rows x 5 columns]
```

In [2]:
```
dataset=dataset.drop(columns=["User ID"])
dataset
```

Out[2]:

	Age	EstimatedSalary	Purchased	Gender_Male
0	19	19000	0	True
1	35	20000	0	True
2	26	43000	0	False
3	27	57000	0	False
4	19	76000	0	True
...	...	...	...	...
395	46	41000	1	False
396	51	23000	1	True
397	50	20000	1	False
398	36	33000	0	True
399	49	36000	1	False

400 rows × 4 columns

In [15]:
```
x=dataset[["Age","EstimatedSalary","Gender_Male"]]
y=dataset[["Purchased"]]
print(y)
dataset.shape
dataset[["Purchased"]].value_counts()
```

```
     Purchased
0            0
1            0
2            0
3            0
4            0
..         ...
395          1
396          1
397          1
398          0
399          1

[400 rows x 1 columns]
```

Out[15]:

```
Purchased
0            257
1            143
Name: count, dtype: int64
```

In [16]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.20,random_state=42)
```

In [18]:
```
from sklearn.linear_model import LogisticRegression
reg=LogisticRegression(penalty='deprecated',C=1.0, l1_ratio=0.0, dual=False,
                       tol=0.0001, fit_intercept=True, intercept_scaling=1,
                       class_weight=None, random_state=0,
                       solver='lbfgs', max_iter=100, verbose=0,
                       n_jobs=-1)
reg.fit(x_train,y_train)
```

```
C:\Users\User\anaconda3\Lib\site-packages\sklearn\linear_model\_logistic.py:1457: FutureWarning:
'n_jobs' has no effect since 1.8 and will be removed in 1.10. You provided 'n_jobs=-1', please
leave it unspecified.
  warnings.warn(msg, category=FutureWarning)
C:\Users\User\anaconda3\Lib\site-packages\sklearn\utils\validation.py:1365: DataConversionWarning:
A column-vector y was passed when a 1d array was expected. Please change the shape of y to
(n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

Out[18]:

▾ LogisticRegression

▸Parameters

▸Fitted attributes

In [19]:
```
y_pred=reg.predict(x_test)
print(y_pred)
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_pred,y_test)
print(cm)
```

```
[0 1 0 1 0 0 1 0 0 0 0 1 0 0 0 0 1 1 0 1 0 0 0 1 0 1 1 0 1 0 0 0 1 0 1 0 0
 0 0 0 0 0 0 0 0 1 0 0 1 0 0 1 0 0 0 0 1 0 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
 0 0 1 1 0 0]
[[50  7]
 [ 2 21]]
```

In [20]:
```
from sklearn.metrics import classification_report
cr=classification_report(y_pred,y_test)
print(cr)
```

```
              precision    recall  f1-score   support

           0       0.96      0.88      0.92        57
           1       0.75      0.91      0.82        23

    accuracy                           0.89        80
   macro avg       0.86      0.90      0.87        80
weighted avg       0.90      0.89      0.89        80
```

In [21]:
```
from sklearn.metrics import r2_score
r=r2_score(y_pred,y_test)
print(r)
```

```
0.45080091533180766
```

In [ ]:

In [ ]: